Chapter 13

Simple Interest

When a person borrows some amount of money from another person or organization (bank), then the person borrowing money (borrower) pays some extra money during repayment, that extra money during repayment is called **Interest**.
For example: If A takes ₹50 from B and after using ₹50, A returns ₹55 to B, then A pays (55 - 50) *i.e.,* ₹5 as interest.

Let us consider following definitions before proceeding chapter

Principal (P): Principal is the money borrowed or deposited for a certain time.

Amount (A): The sum of principal and interest is called amount.

Amount = Principal + Simple Interest

Rate of Interest (R): It is the rate at which the interest is charged on principal. It is always specified in percentage terms.

Time (T): The period, for which the money is borrowed or deposited, is called time.

13.1 Simple Interest (SI)

If the interest is calculated on the original principal for any length of time, then it is called **simple interest**.

$$\textbf{Simple Interest} = \frac{\text{Principle} \times \text{Rate} \times \text{Time}}{100}$$

OR

$$\textbf{SI} = \frac{P \times R \times T}{100}$$

Basic Formulae Related to Simple Interest

$$P = \frac{100 \times A}{100 + RT}$$
$$SI = \frac{ART}{100 + RT}$$

where, SI = Simple Interest, P = Principal, R = Rate of interest, T = Time, A = Amount

Ex. 1 Find the simple interest on ₹200 for 5 yr at 6% per annum.

Sol. Here, $P = ₹\,200,\ T = 5\ yr, R = 6\%$

$$\therefore \quad SI = \frac{P \times R \times T}{100} = \frac{200 \times 5 \times 6}{100} = ₹\,60$$

Ex. 2 In what time, ₹1200 will become ₹1450 when annual rate of interest is 20%?

Sol. Here, $P = ₹\,1200,\ A = ₹\,1450,\ R = 20\%$

We know that, $A = P + SI$

$$\Longrightarrow \quad 1450 = 1200 + SI$$
$$\Longrightarrow \quad SI = 1450 - 1200 = ₹\,250$$

Again, $SI = \frac{P \times R \times T}{100}$

$$\Longrightarrow \quad 250 = \frac{1200 \times 20 \times T}{100} = 240T$$
$$\therefore \quad T = \frac{25}{24} = 1\frac{1}{24} \text{ yr}$$

Ex. 3 A sum at simple interest of 4% per annum amounts to ₹3120 in 5 yr. Find the sum.

Sol. Here, $T = 5\ yr,\ R = 4\%,\ A = ₹3120$

We know that, $P = \frac{100 \times A}{100 + RT}$

$$= \frac{100 \times 3120}{100 + 4 \times 5} = \frac{100 \times 3120}{120} = ₹\,2600$$

Alternate Method

Let sum be P,

Then, $SI = \frac{P \times R \times T}{100} = \frac{P \times 4 \times 5}{100} = \frac{P}{5}$

$\therefore$ Amount $= P + SI = P + \frac{P}{5} = \frac{6P}{5}$

According to the question

$$\text{Amount} = 3120 \implies \frac{6P}{5} = 3120$$

$$\therefore \quad P = \frac{3120 \times 5}{6} = ₹2600$$

Ex. 4 A takes ₹3000 from B for 2 yr at the rate of 10% half-yearly interest. What amount will be paid by A to B after the end of 2 yr?

Sol. Here, $P = ₹3000$, $T = 2$ yr $= 4$ half-yr, $R = 10\%$per annum $= \frac{10}{2}\%$half-yearly

$$\therefore \quad SI = \frac{P \times R \times T}{100} = \frac{3000 \times 4 \times 10}{100 \times 2} = ₹600$$

Amount paid to B $= P + SI = 3000 + 600 = ₹3600$

NOTE

1. If rate of interest is half-yearly, then rate $= \left(\frac{R}{2}\right)$ and time $= 2T$
2. If rate of interest is quaterly, then rate $= \left(\frac{R}{4}\right)$ and time $= 4T$
3. If rate of interest is monthly, then rate $= \left(\frac{R}{12}\right)$ and time $= 12T$
4. To calculate interest, the day on which amount is deposited, is not counted but the day on which amount is withdrawn, is counted.

Ex. 5 Amit takes some loan from Akash for 2 yr at the rate of 5% per annum and after 2 yr he gave back ₹6600 to Akash and completed the payment of his loan. Find the interest paid by Amit.

Sol. Here, $T = 2$ yr, $R = 5\%$, $A = ₹6600$

We know that

$$SI = \frac{ART}{100 + RT} = \frac{6600 \times 5 \times 2}{100 + 5 \times 2}$$
$$= \frac{6600 \times 10}{110} = ₹600$$

Alternate Method

$$SI = \frac{P \times R \times T}{100} = \frac{P \times 5 \times 2}{100} = \frac{P}{100}$$

We know that, $A = P + SI$

$$P + \frac{P}{10} = 6800$$

$$\implies \quad \frac{11P}{10} = 6600$$

$$\therefore \qquad P = \frac{6600 \times 10}{11} = 6000$$

$\therefore$ Interest Paid=Amount-Principal=6600-6000=₹600

13.2 Instalments

When a borrower paid the total money in some equal parts (*i.e.,* not in a single amount), then we say that he/she is paying in instalments.

For example: A borrowed ₹100 from B and he pays back it to B in several parts *i.e.,* ₹20 in 5 times or ₹50 in 2 times etc.
The important point is that borrower has to also pay the interest for using the borrowed sum or purchased article.
In general, the value of each instalment is kept constant even when the interest charged on each instalment vary for each instalment for n equal instalments we only calculate upto $(n-1)$ term.
For simple interest

$$A = \left[x + \left(x + \frac{x \times R \times 1}{100}\right) + \left(x + \frac{x \times R \times 2}{100}\right) + \left(x + \frac{x \times R \times 3}{100}\right) + \ldots\right]$$

where, A = Total amount paid x = Value of each instalment.

Also

$$A = P + \frac{P \times n \times R}{100}$$

where, P is the principal, n is the number of instalments and R is the rate of interest

Ex. 6 A scooty is sold by an automobile agency for ₹19200 cash or for ₹4800 cash down payment together with five equal monthly instalments. If the rate of interest charged by the company is 12% per annum, then find each instalment.

Sol. Balance of the price to be paid through instalments

$$P = 19200 - 4800 = 14400$$

Now, according to the formula

$$A = \left[x + \left(x + \frac{x \times R \times 1}{100}\right) + \left(x + \frac{x \times R \times 2}{100}\right) + \left(x + \frac{x \times R \times 3}{100}\right) + \ldots\right]$$

where, $A = P + \dfrac{P \times n \times R}{100}$

$$\implies 14400 + \frac{14400 \times 12 \times 5}{100 \times 12} = \left[x + \left(x + \frac{12x}{100 \times 12}\right) + \left(x + \frac{12x \times 2}{100 \times 12}\right) + \left(x + \frac{12x \times 3}{100 \times 12}\right) + \left(x + \frac{12x \times 4}{100 \times 12}\right)\right]$$

$$\implies \qquad 14400 + 144 \times 5 = 5x + \left(\frac{x}{100} + \frac{2x}{100} + \frac{3x}{100} + \frac{4x}{100}\right)$$

$$\implies \qquad 15120 = 5x + \frac{x}{10}$$

$$\implies \qquad 15120 = \frac{51x}{10}$$

$$\implies \qquad x = \frac{15120 \times 10}{51} = ₹2964.70$$

Note: In the left hand side and right hand side, given amounts are equal. Each amount is equal to the total amount payable after 5 months.

13.3 Fast Track Techniques to solve the QUESTIONS

Technique 1

If a sum of money becomes n times in T yr at simple interest, then formula for calculating rate of interest will be given as

$$R = \frac{100(n-1)}{T}\%$$

Ex. 7 A sum of money becomes four times in 20 yr at SI. Find the rate of interest.

Sol. Given, T = 20 yr, Let sum = P

Then, the sum after 20 yr = $4P$

$$SI = 4P - P = 3P$$

Now, $$3P = \frac{PRT}{100} = \frac{P \times R \times 20}{100}$$

$$\implies \qquad 3 = \frac{20R}{100} = \frac{R}{5}$$

$$\therefore \qquad R = 15\%$$

Fast Track Method

Here, $T = 20$ yr , $n = 4$

$$\therefore \qquad R = \frac{100(n-1)}{T} = \frac{100(4-1)}{20} = \frac{100 \times 3}{20} = 15\%$$

Technique 2

(i) If a sum of money at a certain rate of interest becomes n times in T_1 yr and m times in T_2 yr, then formula for T_2 will be given as

$$T_2 = \left(\frac{m-1}{n-1}\right) \times T_1$$

(ii) If a sum of money in a certain time becomes n times at R_1, rate of interest and m times at R_2 rate of interest, then formula for R2 will be given as

$$R_2 = \left(\frac{m-1}{n-1}\right) \times R_1$$

Ex. 8 A sum becomes two times in 5 yr at a certain rate of interest. Find the time in which the same amount will be 8 times at the same rate of interest.

Sol. Let Sum = P

Then, for 5 yr,

$$SI = 2P - P = P$$

$$\because \quad SI = \frac{P \times R \times T}{100}$$

$$\therefore \quad P = \frac{P \times R \times 5}{100} = \frac{PR}{20}$$

$$\Longrightarrow \quad R = 20\%$$

Again, for another time(T),

$$SI = 8P - P = 7P$$

$$\therefore \quad 7P = \frac{P \times 20 \times T}{100} = \frac{TP}{5}$$

$$\therefore \quad T = 7 \times 5 = 35 \text{ yr}$$

Fast Track Method

Here, $n = 2, m = 8, T_1 = 5, T_2 = ?$

$$\therefore \quad T_2 = \left(\frac{m-1}{n-1}\right) \times T_1 = \left(\frac{8-1}{2-1}\right) \times 5 = 35\text{yr}$$

Ex. 9 In a certain time, a sum becomes 3 times at the rate of 5% per annum. At what rate of interest, the same sum becomes 6 times in same duration?

Sol. Let Sum = P

Then, for 5 % rate of interest,

$$SI = 3P - P = 2P$$

$$\because \quad SI = \frac{P \times R \times T}{100}$$

$$\therefore \quad 2P = \frac{P \times 5 \times T}{100} = \frac{PT}{20}$$

$$\Longrightarrow \quad T = 40 \text{ yr}$$

Again, for another Rate%(R),

$$SI = 6P - P = 5P$$

$$\therefore \quad 5P = \frac{P \times R \times 40}{100} = \frac{2PR}{5}$$

$$\therefore \quad R = \frac{25}{2} = 12.5\%$$

Fast Track Method

Here, $n = 3, m = 6, R_1 = 5\%, R_2 = ?$

$$\therefore \quad R_2 = \left(\frac{m-1}{n-1}\right) \times R_1 = \left(\frac{6-1}{3-1}\right) \times 5 = \frac{25}{2} = 12/5\%$$

Technique 3

If a certain sum P in a certain time amounts to ₹A_1, at the rate of $R_1\%$ and the same sum amounts to ₹A_2 at the rate of $R_2\%$, then

$$P = \left(\frac{A_2R_1 - A_1R_2}{R_1 - R_2}\right) \quad \text{and} \quad T = \left(\frac{A_1 - A_2}{A_2R_1 - A_1R_2}\right) \times 100$$

Note: If the rate of interest is uniform in above mentioned condition and time is variable, then

$$P = \left(\frac{A_2T_1 - A_1T_2}{T_1 - T_2}\right) \quad \text{and} \quad R = \left(\frac{A_1 - A_2}{A_2T_1 - A_1T_2}\right) \times 100$$

where, T_1 and T_2 are the time for first and second conditions, respectively Ex. 10 A certain sum in certain time becomes ₹500 at the rate of 8% per annum SI and the same sum amounts to ₹200 at the rate of 2% per annum SI in the same duration. Find the sum and time.

Sol. According to the question,

$$A_1 - A_2 = 500 - 200$$

$$\left(P + \frac{P \times 8 \times T}{100}\right) - \left(P + \frac{P \times 2 \times T}{100}\right) = 500 - 200$$

$$\implies \quad \frac{8PT}{100} - \frac{6PT}{100} = 300 \implies \frac{6PT}{100} = 300$$

$$\implies \quad PT = \frac{300 \times 100}{6} = 5000$$

Again, for 8% rate, $SI = \frac{PTR}{100} = \frac{5000 \times 8}{100}$ [as PT=5000]

$$\implies \quad SI = ₹400$$

$$\therefore \quad \text{Sum } (P) = 500 - 400 = 100 \quad \text{[as P=A-SI]}$$

We have, $PT = 5000$

$$\therefore \quad T = \frac{5000}{P} = \frac{5000}{100} = 50 \text{ yr}$$

Fast Track Method

Here, $R_1 = 8\%,\ R_2 = 2\%,\ A_1 = ₹\,500,\ A_2 = ₹\,200$

Now, according to the formula,

$$P = \left(\frac{A_2R_1 - A_1R_2}{R_1 - R_2}\right)$$

$$= \left(\frac{200 \times 8 - 500 \times 2}{8 - 2}\right) = \frac{1600 - 1000}{6} = \frac{600}{6} = ₹100$$

$$\text{and Time, } T = \left(\frac{A_1 - A_2}{A_2R_1 - A_1R_2}\right) \times 100$$

$$= \left(\frac{500 - 200}{200 \times 8 - 500 \times 2}\right) \times 100 = \frac{300}{600} \times 100 = 50 \text{ yr}$$

Ex. 11 A certain sum at a certain rate of SI amounts to ? 1125 in 4 yr and ? 1200 in 7 yr. Find the sum and rate of interest.

Sol. According to the question,

$$\left(P + \frac{P \times R \times 7}{100}\right) - \left(P + \frac{P \times R \times 4}{100}\right) = 1200 - 1125$$

$$\implies \frac{7PR}{100} - \frac{4PR}{100} = 75 \implies \frac{3PR}{100} = 75$$

$$\implies PR = \frac{75 \times 100}{3} = 2500$$

$$\text{For 4 yr,} \quad SI = \frac{PTR}{100} = \frac{2500 \times 4}{100} = ₹100$$

$$\therefore \quad \text{Sum } (P) = 1125 - 100 = ₹1025$$

$$\text{Again, we have, } PR = 2500$$

$$\therefore \quad R = \frac{2500}{1025} = 2.43\%$$

Fast Track Method

Here, $T_1 = 4\ yr,\ T_2 = 7\ yr,\ A_1 = ₹\,1125,\ A_2 = ₹\,1200$

According to the formula,

$$P = \left(\frac{A_2T_1 - A_1T_2}{T_1 - T_2}\right) = \left(\frac{1200 \times 4 - 1125 \times 7}{4 - 7}\right)$$

$$= \frac{4800 - 7875}{-3} = \frac{-3075}{-3} = ₹1025$$

$$\text{We know that,} \quad R = \left(\frac{A_1 - A_2}{A_2T_1 - A_1T_2}\right) \times 100 = \left(\frac{1125 - 1200}{1200 \times 4 - 1125 \times 7}\right) \times 100$$

$$= \frac{-7500}{4800 - 7875} = \frac{-7500}{-3075} = 2.43\%$$

Technique 4

If SI for a certain sum P_1, for time T_1 and rate of interest R_1, is I_1, and SI for another sum P_2 for time T_2 and rate of interest R_2 is I_2, then

$$\text{Difference of } SI = I_2 - I_1 = \frac{P_2R_2T_2 - P_1R_1T_1}{100}$$

1. In above mentioned condition, if all the parameters are constant but time is variable, then

$$I_2 - I_1 = \frac{PR(T_2 - T_1)}{100}$$

2. When only rate of interest is variable, then

$$I_2 - I_1 = \frac{PT(R_2 - R_1)}{100}$$

3. When only sum is variable then

$$I_2 - I_1 = \frac{RT(P_2 - P_1)}{100}$$

4. When only one parameter remains constant and the remaining are variables, then

$(a)\quad I_2 - I_1 = \dfrac{P(R_2T_2 - R_1T_1)}{100}$ $\quad (b)\quad I_2 - I_1 = \dfrac{R(P_2T_2 - P_1T_1)}{100}$

$(c)\quad I_2 - I_1 = \dfrac{T(P_2R_2 - P_1R_1)}{100}$

Ex. 12 Simple interest for the sum of ₹1500 is ₹50 in 4 yr and ₹80 in 8 yr. Find the rate of SI.

Sol. Given that $I_1 = 50,\ I_2 = 80,\ T_1 = 4\ yr,\ T_2 = 8\ yr,\ P =$ ₹1500

According to the question,

$$\frac{P_1 \times R \times T_1}{100} - \frac{P_2 \times R \times T_2}{100} = I_2 - I_1$$

$$\frac{1500 \times R \times 8}{100} - \frac{1500 \times R \times 4}{100} = 80 - 50$$

$$\implies \frac{6000R}{100} = 30$$

$$\therefore \quad R = \frac{30 \times 100}{6000} = \frac{1}{2} = 0.5\%$$

Fast Track Method

Here, $I_1 = 50,\ I_2 = 80,\ T_1 = 4\ yr,\ T_2 = 8\ yr,\ P =$ ₹1500

According to the formula,

$$I_2 - I_1 = \frac{PR(T_2 - T_1)}{100}$$

$$\Rightarrow \quad 80 - 50 = \frac{1500 \times R(8-4)}{100}$$

$$\Rightarrow \quad 30 = 15 \times R \times 4$$

$$\Rightarrow \quad 1 = 2R \Rightarrow R = \frac{1}{2}$$

$$\therefore \quad R = 0.5\%$$

Ex. 13 Simple interest for a sum of ₹1550 for 2 yr is ₹20 more than the simple interest for ₹1450 for the same duration. Find the rate of interest.

Sol. Given that $P_1 = ₹1550,\ T_1 = 2, P_2 = ₹1450,\ T_2 = 2$

According to the question,

$$\frac{P_1 \times R \times T_1}{100} - \frac{P_2 \times R \times T_2}{100} = 20$$

$$\frac{1550 \times R \times 2}{100} - \frac{1450 \times R \times 2}{100} = 20$$

$$\Rightarrow \quad \frac{200R}{100} = 20$$

$$\therefore \quad R = \frac{20 \times 100}{200} = 10\%$$

Fast Track Method

Here, $I_1 - I_2 = ₹20,\ P_1 = ₹1550,\ P_2 = ₹1450,\ T = 2\ yr$

According to the formula,

$$I_2 - I_1 = \frac{RT(P_2 - P_1)}{100}$$

$$\Rightarrow \quad -20 = \frac{R \times 2(1450 - 1550)}{100} \qquad [\because I_2 - I_1 = -I_1 - I_2 = -20]$$

$$\Rightarrow \quad -10 = \frac{-100R}{100}$$

$$\therefore \quad R = 10\%$$

Ex. 14 For a certain sum, the simple interest in 2 yr at 8% per annum is ₹90 more than the simple interest in 1.5 yr at the rate of 10% per annum for the same sum. Find the sum.

Sol. According to the question,

$$\frac{P \times 8 \times 2}{100} - \frac{P \times 10 \times 1.5}{100} = 90 \Rightarrow \frac{P}{100} = 90$$

$\therefore \quad P = ₹9000$

Fast Track Method

Here, $I_1 - I_2 = ₹90,\ T_1 = 2\ yr,\ R_1 = 8\%,\ R_2 = 10\%,\ T_2 = 1.5\ yr$

According to the formula,

$$I_2 - I_1 = \frac{P(R_2T_2 - R_1T_1)}{100}$$

$$\Longrightarrow \quad -90 = \frac{P}{100}(10 \times 1.5 - 8 \times 2)$$

$$\Longrightarrow \quad P = \frac{-90 \times 100}{15 - 18} = ₹9000$$

Technique 5

If $\frac{1}{x}$ part of a certain sum P is lent out at $R_1\%\ SI$, $\frac{1}{y}$ part is lent out at $R_2\%\ SI$ and the remaining $\frac{1}{z}$ part at $R_3\%\ SI$ and this way the interest received be I, then

$$P = \frac{I \times 100}{\frac{R_1}{x} + \frac{R_2}{y} + \frac{R_3}{z}}$$

Ex. 15 Alok lent out a certain sum. He lent 1/3 part of his sum at 7% SI, 1/4 part at 8% SI and remaining part at 10% SI. If ₹510 is his total interest, then find the money lent out.

Sol. Let entire sum=P

Now, according to the question,

$$\frac{1}{3}P \times 7\% + \frac{1}{4}P \times 6\% + \left[1 - \left(\frac{1}{3} + \frac{1}{4}\right)\right] \times P \times 10\% = 510$$

$$\Longrightarrow \quad \frac{\frac{P}{3} \times 7}{100} + \frac{\frac{P}{4} \times 8}{100} + \frac{\frac{5P}{12} \times 10}{100} = 510$$

$$\Longrightarrow \quad \frac{7P}{3} + 2P + \frac{25P}{6} = 510 \times 100$$

$$\therefore \quad P = \frac{510 \times 100 \times 6}{51} = ₹6000$$

Fast Track Method

Here, $R_1 = 7\%,\ R_2 = 8\%,\ R_3 = 10\%$

and $\frac{1}{x} = \frac{1}{3},\ \frac{1}{y} = \frac{1}{4},\ I = ₹510$

$$\therefore \qquad \frac{1}{z} = \left[1 - \left(\frac{1}{3} + \frac{1}{4}\right)\right] = \frac{5}{12}$$

According to the formula,

$$P = \frac{I \times 100}{\frac{R_1}{x} + \frac{R_2}{y} + \frac{R_3}{z}} = \frac{510 \times 100}{\frac{7}{3} + \frac{8}{4} + \frac{50}{12}}$$

$$= \frac{51000}{\frac{7}{3} + 2 + \frac{25}{6}} = \frac{51000}{51} \times 6 = ₹6000$$

Technique 6

A sum of ₹P is lent out in n parts in such a way that the interest on first part at R_1% for T_1 yr, the interest on second part at R_2% for T_2 yr and the interest on third part at R_3% for T_3 yr and so on are equal, then the ratio in which the sum was divided in n parts is given by

$$\frac{1}{R_1 T_1} : \frac{1}{R_2 T_2} : \frac{1}{R_3 T_3} : \ldots : \frac{1}{R_n T_n}$$

Ex. 16 A sum of ₹7700 is lent out in two parts in such a way that the interest on one part at 20% for 5 yr is equal to that on another part at 9% for 6 yr. Find the two sums.

Sol. Let the first sum be ₹x.

Then, second sum =₹$(7700 - x)$

Now, according to the question,

$$\frac{x \times 20 \times 5}{100} = \frac{(7700 - x) \times 9 \times 6}{100}$$

$$\Longrightarrow \qquad 50x = (7700 - x) \times 27$$

$$\Longrightarrow \qquad 50x = 7700 \times 27 - 27x$$

$$\Longrightarrow \qquad 77x = 7700 \times 27$$

$$\Longrightarrow \qquad \text{First sum } x = ₹\ 2700$$

$$\therefore \qquad \text{The second part} = (7700 - 2700) = ₹\ 5000$$

Fast Track Method

Here, Sum $= 7700,\ R_2 = 20\%,\ T_1 = 5\ yr,\ R_2 = 9\%,\ T_2 = 6\ yr$

$$\text{Ratio of two sums} = \frac{1}{R_1 T_1} : \frac{1}{R_2 T_2} = \frac{1}{20 \times 5} : \frac{1}{9 \times 6}$$

$$= \frac{1}{10 \times 5} : \frac{1}{9 \times 3} = \frac{1}{50} : \frac{1}{27}$$

$$= 27 : 50$$

$$\therefore \quad \text{First Part} = \frac{27}{27+50} \times 7700 = \frac{22}{77} \times 7700 = ₹\,2700$$
$$\text{Second Part} = \frac{50}{27+50} \times 7700 = \frac{50}{77} \times 7700 = ₹\,5000$$

Technique 7

The annual payment that will discharge a debt of ₹P due in T yr at the rate of interest R% per annum is given by

$$\text{Annual Payment} = \frac{100P}{100T + \dfrac{RT(T-1)}{2}}$$

Ex. 17 What annual payment will discharge a debt of ₹848 in 8 yr at 8% per annum?

Sol. Here, $P = ₹\,848,\ T = 8\ yr,\ R = 8\%$

According to the formula.

$$\text{Annual Payment} = \frac{100P}{100T + \dfrac{RT(T-1)}{2}}$$
$$= \frac{100 \times 848}{100 \times 8 + \dfrac{8 \times 8(8-1)}{2}}$$
$$= \frac{848 \times 100}{800 + 32 \times 7} = \frac{84800}{1024} = ₹\,82.8125$$